Software Controller Solution Document

The software stack behind the tree: what each tool does, how the controller receives its sequences, how it's turned on/off from software, and how to commission it once per season.

See also: README · Single Page Explanation · Technical Project Document · Setup (Day-of)

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1. Software stack overview

Control happens as a seamless two-part process: **xLights designs on the laptop**, and **WLED plays back on the controller**. A phone provides everyday control.

Layer	Tool	Runs on	Role
Design & sequencing	xLights	Laptop (Win/Mac)	Build the 3D cone model, author timeline effects, output raw pixel data over the network.
Controller firmware	WLED	ESP32 (on the tree)	Receive network data, translate it to electrical signals, store presets, expose a control API, host the Wi-Fi AP.
Daily control	WLED Mobile App / Web UI	Phone	On/off, brightness, speed, select saved playlists — no laptop needed.



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2. What is xLights?

xLights is **free, open-source layout and sequencing software** that runs on your laptop — essentially a visual sequencer, like video-editing software or a digital audio workstation, but built for lights.

- **It understands 3D shapes.** A tree is a cone, not a flat screen. xLights builds a virtual 3D model of your 12-string cone and handles the maths so waves/spirals map perfectly onto the physical shape without warping at the top.
- **Timeline editor.** Drag-and-drop effects (color waves, falling snow, fireworks, scrolling text) onto a time-stamped grid; stretch/shrink to control speed, repeats and transitions. It calculates the exact color and brightness for each of the 4,320 LEDs for every fraction of a second.

- **It does the heavy lifting.** The laptop renders everything and converts the timeline into a stream of raw network packets pushed to the tree.

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3. What is WLED?

WLED is **free, open-source firmware** flashed onto the ESP32 inside the QuinLED-Dig-Quad. It is the brain permanently attached to the tree.

- **Interpreter.** It listens on the network; when xLights streams data, WLED instantly translates those packets into the electrical signals the WS2815 strips need.
- **Standalone player.** WLED has 100+ built-in effects and can store **presets** and **playlists** in the controller's internal flash storage.
- **Control API.** WLED runs a built-in web server and exposes an HTTP/JSON API, which is what the mobile app and web UI use.

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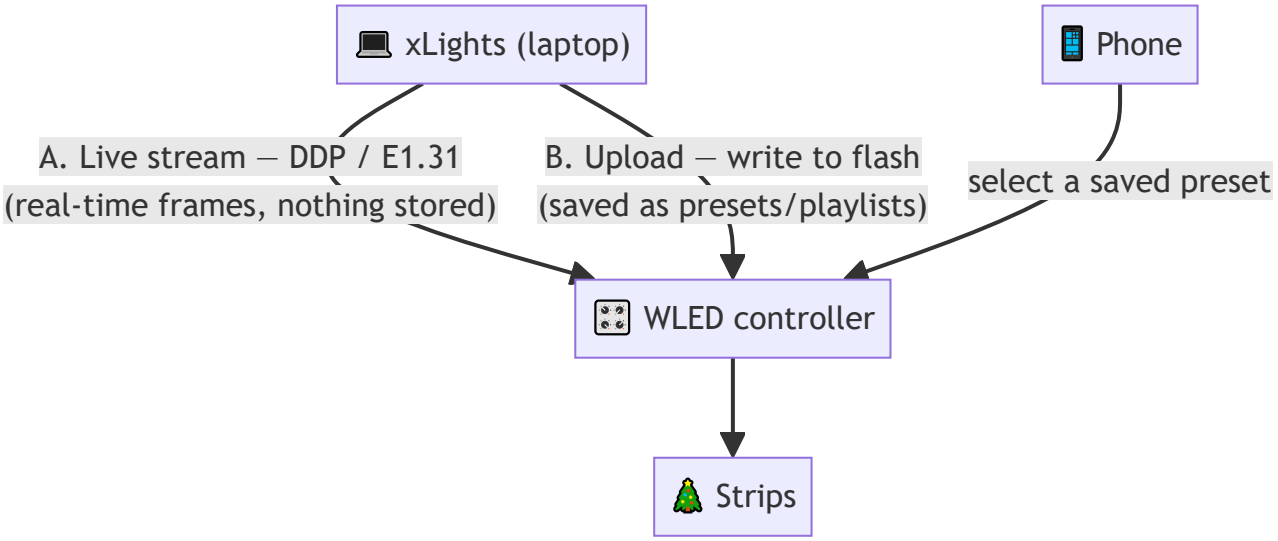
4. How Wi-Fi comes into play

The controller runs in **Standalone Access Point (AP) mode** — it creates its own private, localized Wi-Fi hotspot out in the yard (like a home router, but **not** connected to the internet). Both the laptop and the phone connect directly to this network (e.g. `MatrixTree-WiFi`) to talk to the tree.

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5. How the controller GETS its effects & sequences

There are **two delivery paths** into the controller. Both are used at different times.



A. Live stream (DDP / E1.31)

- xLights pushes rendered frames over Wi-Fi **in real time**; nothing is saved on the controller.
- **DDP (Direct Drive Protocol)** is highly optimized for Wi-Fi and handles thousands of pixels with ease; **E1.31 (sACN)** is the alternative.
- **When used:** while *designing and tuning* — you see laptop edits appear on the physical tree instantly.

B. Uploaded presets / playlists

- Finished sequences are written **once** to WLED's onboard flash storage; the tree then replays them **standalone**, with no laptop attached.
- These saved presets/playlists become the **reusable show library** that you pick from on the day (see Setup).
- **When used:** for normal seasonal operation — upload your finished shows, take the laptop back inside, and run everything from the phone.

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6. How the tree is turned on/off from software

There are two distinct meanings of "off" — and they matter for an always-installed outdoor tree:

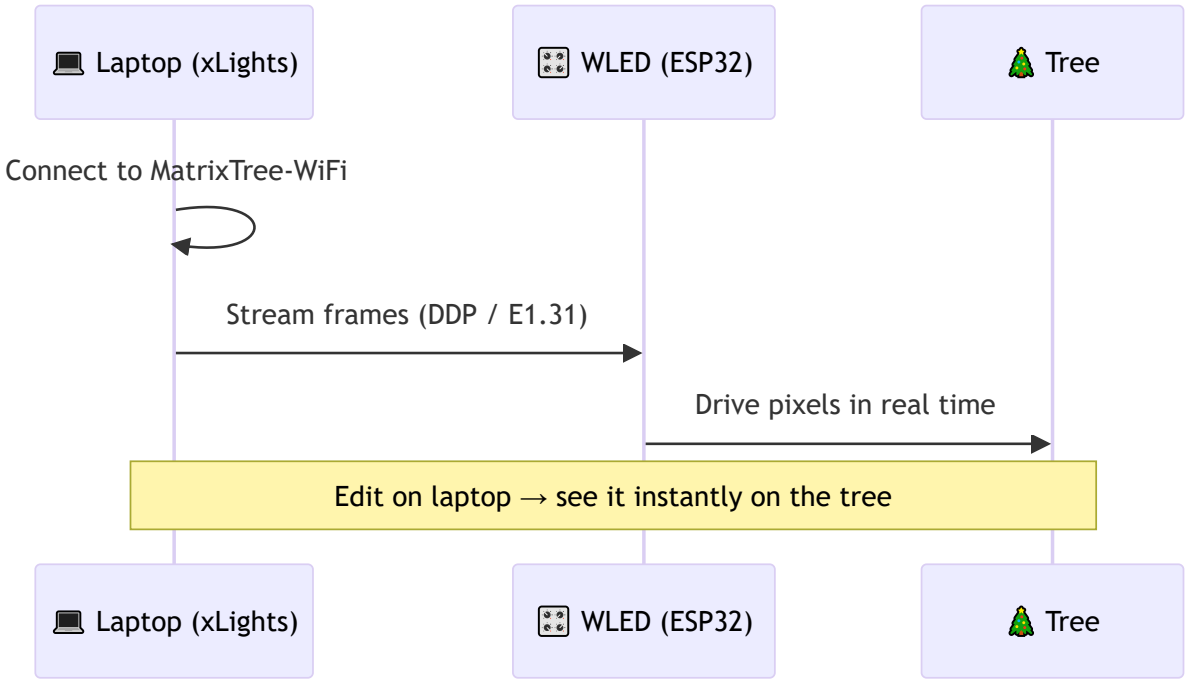
Action	What it does	When you use it
Software on/off (WLED master power toggle / API)	Sets all pixels to black; the ESP32 and PSUs stay energised. Instant, phone-driven.	Normal nightly use — "turn the tree off" at the end of the evening.
True power-off (cut mains to PSUs, e.g. via a smart plug)	De-energises the whole system.	When you want zero standby draw, or before maintenance.

- The **WLED app / web UI** exposes the master on/off toggle plus brightness and effect speed sliders, pause/resume, and playlist selection.
- The same controls are available programmatically via the **WLED HTTP/JSON API** — useful if you ever automate the tree (timers, smart-home triggers).
- An optional **smart plug** on the mains feed gives you the true power-off and can schedule the whole system on/off without touching the software toggle.

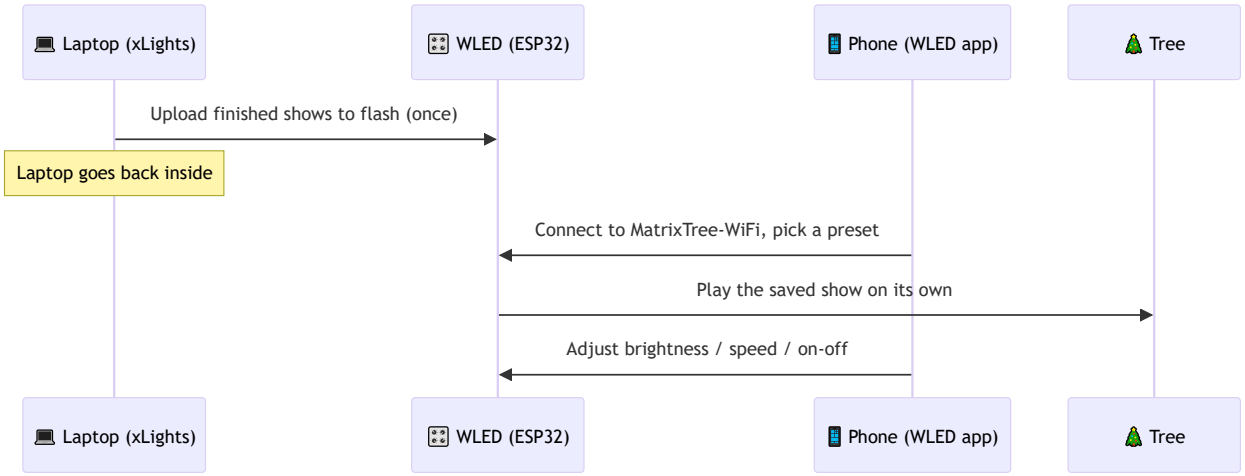
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7. Two operating modes

Mode A — Live stream (for designing)



Mode B — Standalone (for daily use)



8. First-time commissioning (one-time, reusable)

Done **once per season**, not on the day. After this, the controller is prebuilt and reused — day-of operation is covered in Setup.

8.1 Create the WLED standalone Wi-Fi AP

1. Power up the QuinLED-Dig-Quad.
2. Connect a phone/laptop to the default network `WLED-AP` (password `wled1234`).
3. The control panel should auto-open; otherwise browse to `4.3.2.1`.
4. **Config → Wi-Fi Setup → Access Point (AP) settings:**
    - Set a custom **AP SSID** (e.g. `MatrixTree-WiFi`).
    - Set a secure **AP password** (minimum 8 characters).
    - Set the **AP channel** to a clear channel (1, 6 or 11) to avoid neighbourhood interference.
    - Ensure **inbound DDP/E1.31** is **Enabled** (so xLights can stream later).
5. **Save & Reboot** — the controller now broadcasts your custom outdoor network.
6. Configure the **LED Outputs** per Addendum D (GPIO 16/3/1/4, WS281x, 1,080 px each).

8.2 Build the xLights 3D cone model

1. **Controllers tab → Add Ethernet:** name `Main_Tree_Controller`, IP `4.3.2.1`, Protocol **DDP**.
2. **Layout tab:** switch to **3D** view, drop the **Tree** tool onto the canvas.
3. **Properties:** String Type `RGB Nodes` (or `3D`), **# Strings 12**, **Nodes/String 360**, **Spiral Wraps 0** (straight vertical drops).
4. **Wiring direction:** set **Starting Location** (`Top-Left` / `Top-Right`) for straight drops, or use the Custom Model wire map for a zig-zag layout.
5. **Save** the layout — xLights now knows where every one of the 4,320 pixels lives in 3D.

8.3 Build the reusable sequence library

- Author effects/sequences on the xLights timeline.
- Export the compiled shows and **upload them as presets/playlists** to the controller's flash storage.
- These become the library selected on the day.

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9. Why the combination wins

Capability	WLED alone	xLights alone	Combined
Complex 3D matrix effects	❌ (basic 1D)	✅ (3D maths)	xLights designs, WLED displays
Wireless phone control	✅	❌ (needs laptop)	WLED handles daily control
Standalone operation	✅ (flash storage)	❌ (must stay tethered)	xLights exports, WLED plays back

You get unlimited creative design power on the laptop when programming, and effortless, lightweight remote control from the phone the rest of the winter.

Next: Setup — Day-of Operation for running the tree on the day.